

DEPIGMENTING AGENTS

Hyperpigmentation can have a negative psychological impact on a patient's self-esteem, and treatment is therefore critical. Addressing the underlying cause of post-inflammatory hyperpigmentation and maintaining strict photoprotection are essential to prevent further dyschromia. Sun protection is not commonly practiced daily among individuals with ethnically diverse skin types. In many cultures, a deep tan is considered a sign of health. It is important to educate patients with skin of color about the worsening effects of sun exposure on pigmentation disorders.

Hydroquinone-based agents are currently the gold standard for treating hyperpigmentation. Other treatment options include cosmetic camouflage, topical retinoids, azelaic acid, chemical peels, and Q-switched laser therapy. Additionally, several cosmetic "skin brightening" agents—some not approved by the U.S. FDA—are available in the United States, such as **arbutin**, **kojic acid**, **licorice extract**, and **niacinamide**.

Pigmentary (Melanin) Incontinence

Pigmentary or melanin incontinence is a distinct form of post-inflammatory hyperpigmentation that does not respond well to topical depigmenting agents. Histologically, it is characterized by the presence of melanin and melanophages in the upper dermis. Treatment with modalities such as Q-switched laser therapy has shown limited success.

Hydroquinone

Hydroquinone is a hydroxyphenolic compound widely used as a skin-lightening agent. Its primary mechanism of action involves inhibition of the tyrosinase enzyme, thereby blocking the conversion of dihydroxyphenylalanine (DOPA) to melanin.

Additional proposed mechanisms include: - Destruction of melanocytes - Degradation of melanosomes - Inhibition of DNA and RNA synthesis

Hydroquinone is effective in treating various hyperpigmentation disorders, including **melasma**, **post-inflammatory hyperpigmentation**, **lentigines**, and **freckles**. Its efficacy is enhanced when combined with topical retinoids and glucocorticoids.

Formulations and Use

- **Over-the-counter products:** Contain less than 2% hydroquinone
- **Prescription formulations:** Typically 3% to 4% (see Table 224-1)
- **Higher concentrations (5%–10%):** May be compounded for severe cases such as melasma or post-inflammatory hyperpigmentation, but these are more likely to be unstable and irritating

Adjunctive agents such as **vitamin C**, **retinoids**, and **α -hydroxy acids** can improve skin penetration and enhance therapeutic outcomes.

Adverse Effects

- **Irritant and allergic contact dermatitis**
- **Nail discoloration**
- **Post-inflammatory hyperpigmentation** secondary to dermatitis
- **Reversible hypopigmentation**, often presenting as a halo around the treated area—more commonly observed in dark-skinned individuals

To minimize risks, patients should be advised to: - Apply hydroquinone **twice daily** - Limit application to **affected areas only** - Avoid prolonged use

Prolonged use of high-concentration hydroquinone (>6%) over large body surface areas may lead to **exogenous ochronosis**, a bluish-gray discoloration of the skin. This condition is more frequently reported in individuals of African descent and is associated with unregulated or illegal products containing high levels of hydroquinone.

α -HYDROXY ACIDS

α -Hydroxy acids reduce the thickness of the hyperkeratotic stratum corneum by decreasing corneocyte adhesion. They are effective in treating disorders of keratinization, photoaging, and hyperpigmentation. These agents are also used in formulations as cleansers, moisturizers, and chemical peeling agents.

Therapy should begin at low concentrations, with gradual increases in both concentration and frequency of use to minimize side effects.

β -HYDROXY ACID

Salicylic acid is the most commonly used β -hydroxy acid. At concentrations below 6%, it promotes desquamation by causing dyshesion of the stratum corneum, thereby reducing hyperkeratosis. At higher concentrations, salicylic acid becomes cytotoxic and destructive to tissue.

Safety and Use

- **Topical use:** Available in solution form as a superficial peeling agent at 20% and 30% concentrations
- **Efficacy:** Salicylic acid peels are safe and effective, even in darker-skinned patients
- **Systemic toxicity:** Cases of **salicylism** have been reported with widespread, prolonged use—especially in children
- **Local irritation:** Common, but can be minimized by initiating treatment at lower concentrations